Homework #4

For this assignment, you will use attached csv files containing datasets: `NeuronConnect.csv`, `Deflategate.csv`, and `Animal-Weights.csv`.

1. The *C. elegans* connectome has been mapped with exquisite detail, such that we know the number of synapses that connect (nearly) every pair of neurons within *C. elegans*. Your goal is to learn about the distribution of neuron connections within this dataset.

    a. Load the neural connectome data from the csv file. **Plot a histogram of the number of synapses between each pair of neurons in the dataset.**
    b. Use bootstrapping to calculate 95% confidence intervals for the mean and the standard deviation of the number of synapses between each pair of neurons in the dataset. Is this what you expected? **Explain your reasoning and assumptions.**
    c. Is this dataset consistent with one of the four Great Distributions? **Please justify your answer by calculating the necessary statistics, and explain your reasoning.**

2. Some of you may recall the Deflategate controversy from the NFL playoffs several years ago. This homework problem will give you a chance to re-analyze the Deflategate data, which are located in `Deflategate.csv`. Each of the footballs in the dataset belongs to one team (Patriots or Colts) and was measured by two different NFL officials (Blakeman and Prioleau).

    a. Write some code to read the pressure measurements for each football and make a scatterplot of the pressures measured by each official. Color the dots based on which team the footballs came from. **Your output should be a nice-looking scatterplot.**
    b. Are the pressure measurements between officials correlated? Please explain your results. **Your output should include correlation coefficients and a few sentences of explanation.**
    c. For the rest of this problem, we will be using these mean pressures to control for measurement errors. Compute the mean pressure across measurements. **Your output should be a list of mean pressures,**
    d. Compute the means and standard deviation of the mean pressures for each team. **Your output should be a list of means and a list of variances.**
    e. Make a dot plot showing the distribution of pressures by team. You should indicate the mean with a line and add some error bars to the graph. **Please use a sentence or two to explain your choice of error bars.**
    f. Are any of the weight distributions significantly different from one another? Your code should output a list of significant p-values.
    g. Which method(s) did you use for significance testing, and why? **Please use a few sentences to justify each of your choices.**
    h. Discuss two things you would have done differently, if you oversaw the NFL investigation of Deflategate (**a few sentences each**).

3. Scientists have measured the brain weight (in grams) and body weight (in kg) for a variety of different organisms, as documented in `Animal-Weights.csv`. Your goal is to figure out the relationship between these two measurements.

    a. Write some code to read the brain weight and body weight for each animal.
    b. Make scatter plots of body weight vs. brain weight (one plot using linear axes and one plot using logarithmic axes). **Please make the graphs look nice; this is a chance to practice data plotting using Python. We will award bonus points for the prettiest graph in the class.**
    c. Are body weight and brain weight correlated? Calculate correlation coefficients using both Pearson correlation and Spearman correlation. Your answers should include both the correlation coefficients as well as the associated p-values (which should be computed automatically by scipy). **Please write a few sentences to explain your answers.**
    d. Are there any outliers in the data? Can you form any scientific hypotheses related to this? **Please write a few sentences to explain your answers.**